

No argument for fixed base positions: A reply to Heck & Himmelreich (2017)

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Abstract:

In this paper, we discuss Heck & Himmelreich's (2017) analysis of morphological invariant floating quantifiers with a special focus on the question whether their analysis provides an argument in favor of fixed base positions of arguments in German clause structure. We will discuss several shortcomings of Heck & Himmelreich's analysis and will show that their conceptual argument against a base generation approach is based on assumptions that cannot be maintained and that their empirical argument is flawed since it does not take the full range of relevant constructions, and the role of *animacy*, into account. In addition, the present analysis requires less means than Heck & Himmelreich's analysis and can account for data which are very similar to the data discussed in Heck & Himmelreich (2017) but cannot be analyzed in terms of intervention, because there is none. We will thus conclude that Heck & Himmelreich (2017) do not provide an argument for fixed base positions, which is relevant insofar as Salzmann (2025) has recently pointed out that Heck & Himmelreich's argument in favour of fixed base positions stands out against other inconclusive arguments.

1. Introduction

In their analysis of the association between the morphological invariant floating quantifier *alles* 'all' and a *wh*-phrase in German, Heck & Himmelreich (2017: 49, 87) see evidence for a treatment of word order variation among verbal arguments in terms of base positions and a movement analysis, and against variable base generation. A base position approach singles out a specific order as base generated, and derives the other ones via movement, while a variable base generation approach generates different orders directly. The base position approach is the majority approach, but empirical arguments for either approach are hard to find. Recently, Salzmann (2025) evaluated arguments in favour of a base position approach, and found almost all of them inconclusive, Heck & Himmelreich's (2017) being an exception. Arguing from the perspective of a base generation approach with violable linearization constraints (Masloch et al. 2024, 2025), we will evaluate Heck & Himmelreich's analysis.¹ Our conclusion will be that the arguments presented run into empirical problems even if Heck & Himmelreich's assumptions about the floating quantifier *alles* 'all' are granted, but also that a stranding analysis of the morphologically invariant floating quantifier (FQ) is the simpler alternative. The pertinent association between the FQ *alles* and a *wh*-phrase is illustrated in (1).

¹ As Heck & Himmelreich (2017: 85–86) point out, data from the binding of parasitic gaps they discuss does not fit with the conclusions they draw from floating quantifier data in the context of predicates which belong to a minority class with respect to the base order assumed, but they provide an independent reason for this. This reason does not directly extend to the dative-object experiencer-object verbs we will focus on below, and Heck and Himmelreich do not report any tension between floating quantifier and parasitic gaps data for them, so it would be interesting to check parasitic gaps data for the additional cases we discuss below. Since parasitic gaps are generally ungrammatical in our variety of German (all of Heck & Himmelreich's examples involving parasitic gaps are completely unacceptable (to the point of being incomprehensible) to us and to the speakers from our region we showed them to), we will refrain from covering them.

- (1) a. Wer hat euch alles geholfen?
 who.NOM has you.DAT.PL all helped
 'Who all did help you?'
 b. *Die Männer haben euch alles geholfen.
 the.NOM men.NOM have you.DAT.PL all helped
 'The men have all helped you.'

Morphologically invariant FQs require an association with a *wh*-phrase in the same minimal clause, simple NP antecedents are not possible, as is illustrated with (1b).² The FQ must be distinguished from its inflectional kin, see Reis (1992). Since the inflected variant is not the topic of Heck & Himmelreich's analysis, we will ignore it as well. Heck & Himmelreich (2017) claim that the dependence of the acceptability of examples containing FQ *alles* on its positioning relative to the other arguments (well known in the literature, see Pafel 1991) is caused by an opaque intervention effect, namely a blocking of the association between the FQ and the *wh*-phrase by an indefinite. A pertinent example for a verb with a dative object is given in (2) (examples (6a), (7a) from Heck & Himmelreich 2017: 51).

- (2) a. *Wer hat einem Professor alles gratuliert?
 who.NOM has a.DAT professor.DAT all congratulated
 'Who all congratulated a professor?'
 b. Wem hat ein Professor alles gratuliert?
 whom.DAT has a.NOM professor.NOM all congratulated
 'Whom all has a professor congratulated?'

Heck & Himmelreich (2017) use an asterisk to mark (2a) as ungrammatical, while commenting that judgments are subtle (Heck & Himmelreich 2017: 50), and that contrasts are not absolute. Notwithstanding the small differences in acceptability, Heck & Himmelreich present an analysis in which examples like (2b) are well-formed, while examples like (2a) are not. While we agree with Heck & Himmelreich's judgments insofar as relative contrasts are concerned, we do not think that assuming categorical grammaticality contrasts to model them is insightful. Nevertheless, we will continue to use asterisks for the less acceptable variants when discussing Heck & Himmelreich's (2017) theory.

Heck & Himmelreich (2017) propose that the presence of an indefinite NP creates an intervention of the association in (2a), which depends on a prominence hierarchy among the arguments. It disappears if an association is established with a lower argument, but the intervener is a more prominent argument, as is the nominative argument in (2b). The proposed intervention emerges already within *v*P and cannot be accounted for by *wh*-movement. Heck & Himmelreich further propose that the FQ is not a constituent of VP or stranded, but an adverbial that occupies the lowest Spec,*v*. Since an association requires c-command, it can either be established with an element that is base generated in a higher specifier of *v*, or with an element that must move across the FQ from within VP to be re-merged as a higher specifier of *v*. The first possibility is instantiated by the external argument, the second by other arguments of the verb. The latter is also required to satisfy the *Phase Impenetrability Condition* (PIC, Heck & Himmelreich 2017: 58, adapted from Chomsky 2000: 108).

(3) Phase Impenetrability Condition (PIC):

² Reis (1992: 470–471) discusses other possible antecedents, identifying as a minimal requirement that the antecedents must be '*operator phrases that may appear in SpecC position*' (p. 471).

The domain of a head H of a phase HP is not accessible to operations outside of HP. Only HP and its edge domain (comprising of specifiers of H and phrases adjoined to H) are accessible.

A dative or accusative *wh*-phrase originating within VP cannot move into the specifier of C in one fell swoop – this would lead to a violation of the PIC. Instead, an intermediate movement step is required, as is illustrated for (2b) in (4).

(4) $[_{VP} \text{wem}_1 [_{VP} \text{alles} [_{V'} V [_{VP} t_1 \text{gratuliert}]]]]$

The derivation step in (4) does not only satisfy the PIC but also allows an uninterrupted association of the FQ with the *wh*-phrase in terms of a probe and a goal agreeing with one another.³ This is different in (2a). Here, the dative object moves out of VP into Spec,_v, establishing itself in the same position as the *wh*-phrase in (4). Not being a *wh*-phrase, the indefinite object cannot establish an association with the FQ but is able to value the *wh-indefinite* feature of the FQ,¹ thus blocking a complete feature valuation by the *wh*-subject (see (7) below). The nominative *wh*-phrase cannot be realized in a specifier that is higher than the one containing the FQ but lower than the one containing the dative object. This is so because Heck & Himmelreich (2017) resort to the following assumptions:

- (5) Intermediate Step Corollary (Heck & Himmelreich 2017: 59, adapted from Müller 2011: 176):
Intermediate movement steps to specifiers of X triggered by edge features (EFs) must take place before the final specifier is merged within XP.
- (6) Earliness Requirement for Feature Checking (Heck & Himmelreich 2017: 61):
Probe features enter into *Agree* as early as possible.
- (7) Full Match Requirement (Heck & Himmelreich 2017: 67):
A probe on H does not value a goal on H' unless there is a full match between H and H'.

It follows from (5) that the nominative-marked *wh*-phrase cannot be realized before the indefinite object is moved into Spec,_v. It follows from (6) and (7), and from the assumption that the indefinite can value a goal on the FQ that a *partial* agreement relation is established between the indefinite and the FQ, while a *full* match between the *wh*-phrase and the FQ is required according to (7).

Heck & Himmelreich (2017) assume that the nominative-marked external argument is base-generated as last Spec,_v, while dative- and accusative-marked internal arguments are realized within VP. A ditransitive verb thus projects the following base structure (in the simplest case without any movement):

(8) $[_{VP} \text{NP}_{nom} [_{V'} V [_{VP} \text{NP}_{dat} [_{V'} \text{NP}_{acc} V]]]]$

Heck & Himmelreich (2017: 62f.) further assume that multiple movements from VP into Spec,_v are triggered by the same EF, and require a movement-stack. This stack is conceived as a simple push-down automaton, and search of the EF for possible goals proceeds top-down without backtracking. Hence a goal G higher up in the tree is pushed down in the automaton if

³ Our fuzzy formulation here is deliberate: As the discussion in Heck & Himmelreich (2017: 64f.) shows, it is by no means clear which of the two elements is a probe, and which one is a goal.

a lower goal G' is found. Since only the highest element in the stack can re-merge, the order after re-merger in vP will be the same as the order in VP . This can be illustrated for (8) in (9), where $\langle \dots \rangle$ represents the movement-stack (see Heck & Himmelreich 2017: 63):

$$\begin{array}{llll}
 (9) & [_{v'} V[EF] [VP NP_{dat} [_{v'} NP_{acc} V]]] & \Rightarrow & \\
 & [_{v'} V[EF] [VP t_{dat} [_{v'} NP_{acc} V]]] & \langle NP_{dat} \rangle \Rightarrow & \\
 & [_{v'} V[EF] [VP t_{dat} [_{v'} t_{acc} V]]] & \langle NP_{acc}, NP_{dat} \rangle \Rightarrow & \\
 & [_{vP} NP_{acc} [_{v'} V[EF] [VP t_{dat} [_{v'} t_{acc} V]]]] & \langle NP_{dat} \rangle \Rightarrow & \\
 & [_{vP} NP_{dat} [_{vP} NP_{acc} [_{v'} V[EF] [VP t_{dat} [_{v'} t_{acc} V]]]]] & \langle \rangle \Rightarrow & \\
 & [_{vP} NP_{nom} [_{vP} NP_{dat} [_{vP} NP_{acc} [_{v'} V[EF] [VP t_{dat} [_{v'} t_{acc} V]]]]]] & &
 \end{array}$$

Given the condition on the movement-stack, the order between NP_{dat} and NP_{acc} is retained, and the subject, being the final specifier, precedes the other arguments due to the *Intermediate Step Corollary* (5). It is a remarkable aspect of this analysis that putative base positions within VP are deduced from the order outside of VP in addition to the stipulation that the movement-stack does not allow backtracking. For the moment, we will take these assumptions for granted. According to Heck & Himmelreich (2017: 87f.), their analysis also provides an argument against a variable base generation approach: They try to show that “a proponent of the base-generation approach to scrambling” cannot account for (10a, b) and (12) at the same time. They assume that to account for the ungrammaticality of (10a) on a base generation account, one could assume that indefinites realized between the FQ and its antecedent intercept the association between the two.

- (10) a. *Wem hat sie einen Professor alles vorgestellt?
 whom.DAT has she a.ACC professor.ACC all introduced
 b. Wem hat sie alles einen Professor vorgestellt?
 whom.DAT has she all a.ACC professor.ACC introduced
 ‘To whom all did she introduce a professor?’

It follows that the antecedent of the FQ must be realized in adjacency to the FQ. While this assumption is innocuous, if not somewhat unspecific, Heck & Himmelreich’s second supposition is highly problematic.⁴ They state that the indirect object and the direct object can appear in either order. Hence it should be possible to generate the indefinite to the left of the *wh*-object, and the *wh*-object to the left of a FQ (Heck & Himmelreich 2017: 87), which would result in a string like (11a), which can be input to *wh*-movement to Spec,C, resulting in the ungrammatical (10a).

- (11) a. sie einen Professor wem alles vorgestellt hat
 she a.ACC professor.ACC whom.DAT all introduced has
 b. *Wem₁ hat₂ sie einen Professor t₁ alles vorgestellt t₂?
 whom.DAT has she a.ACC professor.ACC all introduced?
 ‘To whom all did she introduce a professor?’

⁴ It should be pointed out here that Heck & Himmelreich (2017) do not refer to a specific base generation proposal. We are not aware of a variable base generation approach that relies on such a condition. As will become clear below, such a condition is not required either. In addition, the concept *adjacency* allows various implementations. Fanselow (2001; 2003) assumes a hierarchical structure of the German vP/VP and still argues for variable base generation, as do Masloch et al. (2024, 2025). Anticipating the analysis proposed below, we would like to point out that adjacency between the *wh*-phrase and the FQ follows from assuming a single constituent of *wh*-phrase and FQ that no other constituent can intervene, because it is outside of the phrase *ipso facto*.

Heck & Himmelreich (2017: 87) continue by arguing that the derivation shown in (11) must be blocked and that the only way to block it would be the requirement that “*the movement path of a wh-phrase associated with a FQ must not cross an indefinite*”. This conclusion, however, is at odds with the grammaticality of (2b) and similar examples with accusative objects, as shown in (12).

- (12) Wen hat ein Professor alles beleidigt?
 who.ACC has a.NOM professor.NOM all insulted
 ‘Who all did a professor insult?’

We will take up this line of reasoning below. As already pointed out, Heck & Himmelreich (2017) do not refer to a specific analysis of word order in terms of variable base generation. We will dispute that base generation approaches must claim that the order of direct and indirect object is arbitrary. Base generation approaches to serialization can refer to linearization constraints based on properties of the phrases (features, to keep it simple), as they are assumed in many works on word order in German and other languages (see the overview in Frey 2015). The linearization constraints will then determine which serializations are acceptable.

In section 3, we will refute the assumption that the ordering of indirect and direct objects is arbitrary, irrespective of the features they carry. Although less depends on the specific implementation of the idea that the *wh*-phrase and the FQ must appear in adjacency, we will for concreteness assume a stranding analysis: the *wh*-phrase and the FQ form a constituent, and the *wh*-phrase may strand the FQ.⁵ Analyses along this line were developed by Pafel (1991) and Reis (1992) and have recently been defended at length by Doliana (2021, 2022). This view allows the formulation of a generalization, which immediately opens an analysis in terms of base generation and linearization constraints. Doliana (2022: 337) provides it as follows: “*Distal alles occurs in any of the positions that its associate can occur in.*” We will return to it in section 3. It should be clear, however, that a consequence can be derived from this assumption, viz. that a FQ may not appear in a position if its associate is banned from the position.

The first argument by Heck & Himmelreich in favour of a fixed base order approach is a conceptual one, but they also present an empirical one. This rests on the assumption that unaccusative verbs taking a nominative and a dative argument and psych verbs with a dative experiencer show a base order in which the nominative argument is merged lower than the dative initially – the dative becomes the final Spec,_v then. Heck & Himmelreich (2017) thus expect a reversal of grammaticality in comparison to (2). For these verbs, the dative cannot act as an antecedent for the FQ in the presence of an intervening nominative, as illustrated in (13) for experiencer-object psych verbs (examples (69b), (69a) in Heck & Himmelreich (2017: 86)):

- (13) a. *?Was für Professoren könnte ein Berufswechsel alles gefallen?
 what for professors.DAT could a.NOM career.change.NOM all appeal.to
 ‘What kind of professors could a career change appeal to?’
 b. Was für Berufswechsel könnten einem Professor alles gefallen?
 what for career.changes.NOM could a.DAT professor.DAT all appeal.to

⁵ To account for the contrasts in (2), (10), and (13), as well as for the possibility that the *wh*-phrase and the FQ may appear together in Spec,C (see (i)), Heck & Himmelreich (2017: 81) have to assume that *alles* may combine with a *wh*-phrase directly, but that stranding is prohibited in this case. Of course, this adds a further complication to their analysis.

(i) [Wen alles]₂ hat sie t₂ beleidigt?
 who.ACC all has she.NOM insulted
 ‘Who all did she insult?’

‘What kind of career changes could appeal to a professor?’

We agree that there is a relative contrast between (13a) and (13b). Heck & Himmelreich, however, do not consider the full set of options. Masloch et al. (2024: 20f.) discuss the results of their Forced Choice Study B with *animate subjects*, which shows that at least for a subset of experiencer-object verbs governing the dative, “*the choices are almost exactly what is expected if participants are forced to choose between two alternatives they like equally well.*” In other words: certain experiencer-object verbs, including *gefallen* (‘appeal to’) show no order preference at all if the subject is animate.⁶ The reason behind this non-preference is the loss of a contrast induced by a difference in animacy between the stimulus subject and the experiencer object. This has of course tremendous consequences for the contrast proposed in Heck & Himmelreich (2017), once examples without a difference in animacy are considered, as can be illustrated in (14):

- (14) a. Was für unabhängige Kandidaten könnten einem enttäuschten Wähler alles
 what for independent candidates.NOM could a.DAT disappointed voter.DAT all
 gefallen?
 appeal.to
 ‘What kind of independent candidates could appeal to a disappointed voter?’
 b. Was für enttäuschten Wählern könnte ein unabhängiger Kandidat
 what for disappointed voters.DAT could a.NOM independent.NOM candidate.NOM
 alles gefallen?
 all appeal.to
 ‘What kind of independent voters could an independent candidate appeal to?’

If the contrast between (13a) und (13b) comes about because of an intervention effect, where the nominative argument is merged lower than the dative argument of *gefallen* (‘appeal to’), the lack of such a contrast in (14) appears to be a mystery. The only relevant difference between (13) and (14) is the animacy of the stimulus, which is a feature of an NP, not of the verb. Even if we ignore this and assume that there are two verbs *gefallen*₁ and *gefallen*₂, which differ regarding the animacy of the stimulus, this will not suffice to account for the lack of a contrast in (14). The only possibility would be to assume that there are not only two (almost completely identical) verbs, but that the arguments of the verb with an animate stimulus can be base generated in either order within VP after all, or that they scramble within VP prior to movement into vP. While these two assumptions appear *ad hoc*, scrambling within VP is a problem that might be addressed in Heck & Himmelreich’s analysis anyway – a point to which we will return below.

Summing up so far, we can observe that the conceptual argument put forth in favour of a base position approach and against a variable base generation approach is incomplete at best and inconclusive at worst. This also hold for the empirical argument, which does not consider the differences between animate and inanimate stimuli.

The remainder of this paper will proceed as follows: In section 2, we will review an inherent problem of the analysis, viz. possible word order variation within VP. In section 3, we briefly summarize the approach by Masloch et al. (2024), which is based on the combination of configurational structures and violable linearization constraints, implemented in a Maximum Entropy Grammar. The grammar allows an implementation of Pafel’s (1991) observation that FQs

⁶ The crucial role of animacy in determining order preferences is well known (see Hoberg 1997, Müller 1999, Heck 2000, inter alia).

may only appear where their associate may. Based on experimental evidence, we will argue that experiencer-object verbs like *gefallen* ('please') are better analysed in allowing variable generation of their arguments.

2. General Assumptions

We agree with Heck & Himmelreich (2017) that an analysis of the contrasts observed in (2), (10), and (13) – but also the lack thereof in (14) – should not be sought in the movement of the *wh*-phrase to Spec,C. The present analysis is agnostic towards the question whether the occupation of Spec,C by a *wh*-phrase – as well as the occupation of C° by V° – requires movement or other means (as have been proposed in Borsley 1989 and Kiss & Wesche 1991). We will take it to be uncontroversial that this operation forms part of the generating component of the grammar (see below).

The constraints that we invoke apply to the structure of the German *Mittelfeld*, just like the constraints employed in Heck & Himmelreich's analysis. What we want to show, however, is that a base generation approach can account for the pertinent contrasts, and that this analysis requires less stipulations, such as conditions on stacks or a *Full Match Requirement*, the application of which makes only sense if a phrase provides a partial match that can never be the antecedent of the FQ. And the proposal will also account for the contrast in (13) and the lack thereof in (14), as well as for contrasts that appear to look like Heck & Himmelreich's data but do not contain an intervener (see the discussion around (23) and (24) below).

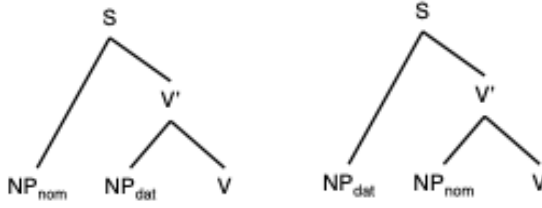
Heck & Himmelreich (2017: 87) seem to suggest that a base generation approach allows all orderings of arguments (and possibly adjuncts), without resorting to further constraints. But approaches relying on base generation usually employ linearization constraints. It is well-known that the order of arguments (and adjuncts) in German clause structure is less restricted than the order in English or other languages, but still that preferred or unmarked orders must be distinguished from marked or dispreferred orders.

The distinction between marked and unmarked, or preferred and dispreferred serializations can be taken to result from the application of linearization constraints, so that these constraints provide similar, if not identical contrasts as the ones observed by Heck & Himmelreich (2017). This holds in particular for the contrast observed in (10), which Heck & Himmelreich (2017) see as an argument against a base generation approach. Now consider the analysis of word order constraints proposed in Masloch et al. (2024, 2025), who employ a *Maximum Entropy Grammar* (MEG; Goldwater & Johnson 2003; Hayes 2022), which is a probabilistic version of Optimality Theory (OT; Prince & Smolensky 2004; Müller 2015).

Just like OT, MEG distinguishes between a *generator function*, which is based on inviolable rules and principles such as the incremental merger of heads and phrases to provide a binary phrasal projection, and a set of constraints that apply to the output of the generator function and determine which of the generated structures is preferred or dispreferred. In such an architecture, ungrammatical structures cannot be derived by the generator function, but violable and weighted constraints provide contrasts in preferences between variants.

To be more specific, we will assume a generator function that creates binary projections from a verb and its arguments. If a verb assumes two arguments and there are no adverbials, we will get two different structures. Only one of the arguments can be realized within a local binary projection, and the generator function leaves it open which of the arguments is discharged first. If a verb has three arguments, we will end at six different structures. The output of the generator function is hence a set consisting of all binary branching trees formed from the input, as is illustrated for our example with two arguments in (15).

(15)



Masloch et al. (2024) assume a set of linearization constraints $C = \{c_1, \dots, c_n\}$ that apply within the maximal projection of a head, so both structures in (15) will receive a violation profile, which determines the probability of a structure to be chosen. The probability of a candidate y , which is an element of the generator output $Y(x)$ is provided in equation (16) (see Goldwater & Johnson 2003: 114):

$$(16) P(y|x) = \frac{\exp(\sum_{i=1}^m w_i f_i(y, x))}{\sum_{y \in Y(x)} \exp(\sum_{i=1}^m w_i f_i(y, x))}$$

The equation in (16) determines the number of violations for each constraint c_i in y given x as $f_i(y, x)$, and multiplies this figure with the weight w_i of c_i . This calculation is summed up for all applicable constraints and exponentiated. Finally, the violation profile for an individual element $y_i \in Y(x)$ is divided by the sum of all violation profiles of all elements $y \in Y(x)$. This last step has the consequence of determining the probability of a generation. We can thus derive probabilities of candidates – and hence contrasts between them – by determining individual violation profiles and comparing them to the summed violation profiles of all structures generated from the same input. This can be illustrated with the examples in (17).

- (17) a. ??Sie meinte, dass [S [NP_{dat} einem Studenten] [V' [NP_{nom} ein Professor] gratuliert hat]].
 she meant that a.DAT student.DAT a.NOM professor.NOM
 gratuliert hat]].
 congratulated has
 b. Sie meinte, dass [[ein Professor] [[einem Studenten] gratuliert hat]].
 'She said that a professor congratulated a student.'

Example (17a) is clearly marked in comparison to (17b) in an out-of-the-blue context or a context where the whole embedded clause is in focus (such as in an answer to the question *Was meinte sie?* 'What did she say?'). We are only interested in the bracketed part of the examples in (17). There are two options to discharge the arguments: nominative first (17a), or dative first (17b). Example (17a) differs from example (17b) in that the former violates the linearization constraint *agent < non-agent*, while the latter does not. In accordance with Masloch et al. (2024), we assume that this constraint takes a weight of -3.6 . The violation profiles of (17a, b) are calculated as follows:

- (18) a. Probability of (17a): $\exp(-3.6 \times 1) / (\exp(-3.6 \times 1) + \exp(0 \times 1)) = 0.027$
 b. Probability of (17b): $\exp(0 \times 1) / (\exp(-3.6 \times 1) + \exp(0 \times 1)) = 0.973$

According to (18) there is a strong contrast between (17a) and (17b).⁷ In (19), we look at the experiencer-object verb *gefallen* ('appeal to'). Following Pesetsky's (1995) distinction between *subject matters* and *causers*, the subject of *gefallen* ('appeal to') is analyzed as a *subject matter*, not a *causer*.

- (19) a. Julia hat erzählt, dass eine Entwicklung einem Besucher gefallen hat.
 Julia has said that a.NOM development.NOM a.DAT visitor.DAT pleased has
 b. Julia hat erzählt, dass einem Besucher eine Entwicklung gefallen hat.
 'Julia said that a development pleased a visitor.'

Since an experiencer-object verb does not introduce an agent, and *gefallen* ('appeal to') specifically does not introduce a causer, there is only one linearization constraint at stake, viz. *animate* < *inanimate*. This constraint shows a weaker weight than the constraint on agency (–1.1):

- (20) a. Probability of (19a): $\exp(-1.1 \times 1) / (\exp(-1.1 \times 1) + \exp(0 \times 1)) = 0.25$
 b. Probability of (19b): $\exp(0 \times 1) / (\exp(-1.1 \times 1) + \exp(0 \times 1)) = 0.75$

The analysis predicts a weaker contrast between (19a, b), if compared to (17a, b). One can then assume that acceptability judgments are closely related to differences in probabilities within the same candidate set. In Ellsiepen and Bader's (2018) discussion of Keller (2000), they conclude that (relative) acceptability judgments are best explained by a model in which degrees of acceptability are proportional to the weights of constraint violations within the same candidate set.

A variable base generation analysis as criticized by Heck & Himmelreich (2017) would be one where all base generated orders are equal. Of course, they are not, and one does not need to assume a fixed order of generation to model this.

3. An alternative analysis: stranding and linearization constraints

A core empirical generalization about the syntactic distribution of morphologically invariant *al-*les goes back to Pafel (1991) and Reis (1992) and has recently been taken up by Doliana (2022: 337): "*Distal alles occurs any of the positions that its associate can occur in*". In the following, we will pursue a particular implementation of this generalization that makes use of the idea that the association between the *wh*-phrase and the FQ is captured by assuming that they form a constituent together, and distal FQs are analysed as being stranded. Given that violable linearization constraints allow to determine relative contrasts by calculating their respective realization probabilities, it follows that a FQ becomes unlikely to appear in a position if its associate is unlikely to appear in the same position.

Before the analysis will proceed, we would like to point out that it differs from Heck & Himmelreich (2017) in a crucial detail: while Heck & Himmelreich (2017: 81) admit that *wh*-phrases and FQs can be combined into single constituents, they also argue that stranding of the FQ cannot be permitted (as already pointed out in fn 6). Without going into details here, it should be clear that a stranding analysis combined with the other assumptions made in Heck & Himmelreich (2017) would seriously undermine their analysis in terms of intervention, and in fact would not allow to derive a contrast in (2), (10), or (13). The present analysis follows Doliana (2022) in assuming a stranding analysis, because we do not rely on an intervention effect, which depends on the interplay of several stipulative considerations. It should thus be kept in

⁷ The corresponding odds of (17a) to (17b) are 1:36. The odds for (19a) to (19b), as calculated in (20), are 1:3, which may provide a more intuitive approach to the strength of the predicted contrasts.

mind that employing a stranding analysis is generally incompatible with an analysis relying on intervention.

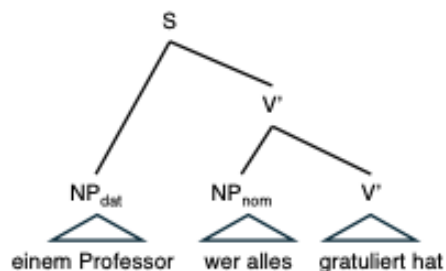
In section 2, we have determined the violation profiles of the serializations in (17) and have concluded that (17a) is highly marked in comparison to (17b), which can be derived from their realization probability. But the subordinate clauses in (17) are sufficiently similar to the examples wielded by Heck & Himmelreich, (2a, b) in particular, which will be repeated here under (21).⁸

- (21) a. ??Wer hat einem Professor alles gratuliert?
 who.NOM has a.DAT professor.DAT all congratulated
 ‘Who all congratulated a professor?’
 b. Wem hat ein Professor alles gratuliert?
 whom.DAT has a.NOM professor.NOM all congratulated
 ‘Whom all has a professor congratulated?’

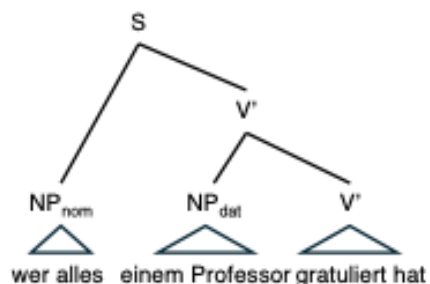
Heck & Himmelreich (2017) assume that an intervention effect is obscured by a subsequent movement of the *wh*-phrase. As already pointed out, we are agnostic towards the analysis of the *wh*-movement into Spec,C, but will assume that prior to movement, the *wh*-phrase and the FQ form a constituent, from which the latter is stranded after movement. We thus agree that the contrast in (21) is to be sought in the domain of the verbal projection, which is subject to the linearization constraints already introduced.

(22)

a.



b.



⁸ As already pointed out, we agree with Heck & Himmelreich (2017) that an account of the observed contrasts should be sought *before* *wh*-movement takes place. It is a consequence of this that constraints determining argument order in the midfield *in general* can be made responsible for the contrasts in (2), (10), and (13).

The generator function generates both (22a) and (22b), and (22a) will differ from (22b) in violating the linearization constraint *agent < non-agent*. If no other linearization constraints are considered, then (22a) and (22b) are in direct competition, and the violation profile of (22b) makes it the unmarked choice in comparison to (22a), which in turn is highly marked, thus accounting for the contrast between the two.⁹

Two consequences follow: First, the analysis predicts that two orders become almost indistinguishable if both orders violate (almost) equally weighted constraints, and secondly, additional constraints may apply in some cases, but not in others, hence yielding changes in relative contrasts again. Let us consider the second point first and look at a case where it becomes particularly clear that treating the acceptability contrasts observed by Heck & Himmelreich in terms of grammaticality differences is not helpful. Word order in German interacts with information structure and prosody in a complex way (see Jacobs 1993, Büring 2016, Truckenbrodt 2017, *inter alia*). A sentence like (17a) is acceptable if the nominative NP is focussed and contains the nuclear pitch accent, as illustrated in (23).

- (23) *Meinte Maya nicht, dass ein Lehrer einem Studenten gratuliert hat?* ‘Didn’t Maya say that a teacher congratulated a student?’
 Sie meinte, dass einem Studenten ein ProFESSor gratuliert hat.
 she meant that a.DAT student.DAT a.NOM professor.NOM congratulated has
 ‘She said that a proFESSor congratulated a student.’

Now consider the examples in (24) with a definite subject, which cannot count as an intervener for Heck & Himmelreich.

- (24) a. Wem hat alles der Professor gratuliert?
 whom.DAT has all the.NOM professor.NOM congratulated
 b. Wem hat der Professor alles gratuliert?
 whom.DAT has the.NOM professor.NOM all congratulated
 ‘Whom all did the professor congratulate?’

While (24b) seems to be preferred when both sentences are presented in isolation, (24a) is perfectly fine in a situation where we ask for an exhaustive list of people who were congratulated by the professor (and not by the student, or the secretary, or whatever the focus alternatives may be).¹⁰ Heck & Himmelreich generally cannot derive sentences like (24a) in which *alles* precedes the subject because the FQ is assumed to be merged as the first specifier of *v*, below the external argument (see Doliana (2021: 52)).¹¹ A linearization-based approach allows to capture the information-structural differences between (24a) and (24b). In a neutral setting,

⁹ Following the arguments presented in Haider (2010), we assume that there are no functional projections between V and C, but the analysis does not depend on this assumption.

¹⁰ FQ *alles* indicates the expectation that the answer contains more than one individual and demands it to be exhaustive (see Reis 1992, Zimmermann 2007).

¹¹ A solution on Heck & Himmelreich’s (2017) account may be to claim that *alles* is scrambled in (24a), i.e. occupies a position to the left of the subject due to further movement in T. However, it is hard to imagine how *alles* could fulfil the information-structural requirements that Heck and Himmelreich (2017:81–83) assume the feature triggering scrambling to Spec,T to impose. Gould (2023:531 sq.) proposes a modification of Heck & Himmelreich’s account in which *alles* merges after the external argument but before any re-merging, which would make place *alles* and the subject in (24a) in base positions, from which the position of the subject in (24b) could be derived by further movement of the subject to Spec,T. The question remains why this movement is not triggered in the context asking for an exhaustive list of people congratulated by the professor, as discussed above.

linearization constraints like *agent* < *non-agent* and *definite* < *indefinite* lead to a preference of (24b), i.e. a linearization where the FQ follows the definite subject. Hence, the serialization in (24a) would lead to a clear decrease in acceptability in comparison to (24b); resulting in a contrast similar to the one observed in (20). When the subject is focused, one may take a linearization constraint *background* < *focus* (Jacobs 1988) to counter *agent* < *non-agent* and *definite* < *indefinite*, so that (24a) is at least as good as (24b) (and probably even preferred, but judgments are subtle).¹²

There are other structures that prove to be even more problematic, because non-subject arguments are realized below the FQ, but in the ‘wrong’ order, as is illustrated in (25).¹³

- (25) a. Wer hat alles einem Zeugen das Foto gezeigt?
 who.NOM has all a.DAT witness.DAT the.ACC foto.ACC showed
 b. Wer hat alles das Foto einem Zeugen gezeigt?
 who.NOM has all the.ACC foto.ACC a.DAT witness.DAT showed
 ‘Who all showed the picture to a witness?’

While (25a) is unproblematic for Heck & Himmelreich’s analysis, the lack of an observable contrast in (25) proves to be a severe problem: both are almost equally acceptable. And the examples show an instance of almost equally weighted linearization constraints that cannot be resolved in the examples given, so that optionality emerges. This line of reasoning will also account for the lack of a contrast in (14), which is unexplained in Heck & Himmelreich (2017). The order in (25a) follows the constraint that indirect objects precede direct objects and animate NPs inanimate ones, while (25b) follows the constraint that definite NPs precede indefinite ones (see Frey 2015, Ellsiepen & Bader 2018). Let us start with the analysis of (25a). The grammaticality of the example does not constitute a problem for Heck & Himmelreich if we assume that the objects remain within the VP, while the association between the *wh*-subject and the FQ takes place within *vP*. Example (25b), however, cannot be derived in Heck & Himmelreich’s system without violating the conditions that allow to determine the contrasts in (2), (10), and (13). If Heck & Himmelreich conceded that the FQ does not occupy the first Spec,*v* by necessity, which would only be a first step towards an analysis of (25b) in blocking the application of the Full Match Requirement, they could not account for the contrasts in (2), (10), and (13) anymore. Even worse is for their proposal is the idea that scrambling can take place within the VP, because scrambling within VP undermines the very idea that the order within *vP* reflects the base order within VP.

While Heck & Himmelreich (2017) explicitly deduce base positions within VP from the idea that movement to Spec,*v* is order-preserving, they also assume implicitly that movement within VP cannot take place. Regarding the very idea that movement from VP to *vP* is order-preserving, this idea goes against most proposals dealing with scrambling in terms of movement. While Heck & Himmelreich (2017) discuss Scandinavian object shift to give their analysis credibility, Haider (2010: 152) – among many others – distinguishes scrambling, understood as re-ordering, from object shift, which keeps the relative order of moved elements constant. Further, order-preservation is not derived from linguistic considerations proper, but from the idea that probes may not backtrack. This stipulation guarantees that goals appear on the movement-stack in the same order as they have appeared within the VP. Heck & Himmelreich (2017: 63)

¹² Linearization constraints along these lines are discussed in Müller (1999) and Frey (2015). The constraint is used here for expository purposes only.

¹³ Example (25) is modelled on a test item from Ellsiepen & Bader’s (2018) study on factors influencing linearization in German, for which both orders were almost equally acceptable (Ellsiepen & Bader 2018:16–17).

state that “*this naturally follows from the top-down manner in which the EF probes into its search space*”. Apparently, they are not aware of search algorithms that allow top-down parallel search *without resorting to backtracking*, as described by Earley (1970). Blocking backtracking is thus by no means a natural or logical consequence of top-down search. Given that this stipulation is problematic, and that movement within VP may be considered, the very idea of a derivation of base positions breaks down.

Having rejected the conceptual argument in favour of base positions in German clause structure, let us now turn the empirical argument again. In (25) we saw a case where conflicting linearization constraints lead to optionality. Another possibility for optionality to arise occurs when no linearization constraint is violated by any candidate. This is what distinguishes (14) from (13), which are repeated here as (26) and (27).

- (26) a. ?Was für Professoren könnte ein Berufswechsel alles gefallen?
 what for professors.DAT could a.NOM career.change.NOM all appeal.to
 ‘What kind of professors could a career change appeal to?’
 b. Was für Berufswechsel könnten einem Professor alles gefallen?
 what for career.changes.NOM could a.DAT professor.DAT all appeal.to
 ‘What kind of career changes could appeal to a professor?’
- (27) a. Was für unabhängige Kandidaten könnten einem enttäuschten
 what for independent candidates.NOM could a.DAT disappointed.DAT
 Wähler alles gefallen?
 voter.DAT all appeal.to
 ‘What kind of independent candidates could appeal to a disappointed voter?’
 b. Was für enttäuschten Wählern könnte ein unabhängiger
 what for disappointed voters.DAT could a.NOM independent.NOM
 Kandidat alles gefallen?
 candidate.NOM all appeal.to?
 ‘What kind of disappointed voters could an independent candidate appeal to?’

The contrast in (26) is provided in Heck & Himmelreich (2017) as evidence for their analysis. How can a base generation approach with linearization constraints account for it? Once again, we will assume following Pafel (1991) that the position of the FQ is indicative of the position of the *wh*-element prior to movement into Spec,C. For (26b), this yields an order complying to the constraint *animate* < *inanimate*, as it was invoked in the analysis of (19b), the subordinate clause of which is structurally identical to (26b) in the relevant parts. This constraint is violated in (26a), as it was in (19a). Example (26b) does not violate any linearization constraint – just like (19b). This accounts for the contrast in (26). The applicable linearization constraints favour the binary branching structure in which the dative NP asymmetrically c-commands the nominative NP, because this structure conforms to the only applicable constraint. The resulting constraint profiles are identical to the ones for (19), as calculated in (20). The lack of a contrast between (27a) and (27b) emerges because the only constraint relevant for (26) cannot be applied in (27): both arguments are animate.

The non-contrast in (27) lacks an explanation in Heck & Himmelreich (2017). As is shown by Masloch et al. (2024) it is in line with a *Maximum Entropy Grammar* that certain candidate sets do not show contrasts at all, if no candidate derived from a given input violates a linearization constraint. Similarly, Kiss et al. (forthcoming) show that violation of two constraints of (nearly) equal weight for two candidates again leads to true optionality in linearization. True optionality in linearization is hard to account for in an analysis assuming base positions, because such an

analysis would have to admit that there are at least some verbs or verb classes that result in more than one base order (reprising a theme that has already been raised in section 2). In addition, one would have to explain how animacy, which is a feature of the elements ordered, not the verb, can make a difference. The contrast in (26) and the lack thereof in (27) can thus be accounted by a base generation approach, but not by the analysis proposed in Heck & Himmelreich (2017).

Finally, let us return to the contrasts in (2) and (10). Although this was already implicit in the discussion in section 2, it should be pointed out explicitly that the analysis of the contrast in (2) is identical to the analysis of the contrast observed in the subordinate clauses in (17), repeated here in (28).

- (28) a. ??Sie meinte, dass [_S [_{NPdat} einem Studenten] [_{V'} [_{NPnom} ein Professor] gratuliert hat]].
 she meant that a.DAT student.DAT a.NOM professor.NOM
 gratuliert hat]].
 congratulated has
 b. Sie meinte, dass [[ein Professor] [[einem Studenten] gratuliert hat]].
 'She said that a professor congratulated a student.'

The contrast in (28) cannot be reduced to an intervention effect for the simple reason that there are no floating quantifiers in the examples. The only additional assumption to be made here – as is in our analysis of morphological invariant FQs in general – is that the *wh*-element and the FQ form a constituent, and that the latter is stranded if the former is realized in Spec,C. Given this, we find structures in (2) prior to movement to Spec,C, and in the subordinate clauses in (28) that are identical in the relevant parts, and the relative contrast in (2) can be accounted for by the same means that the relative contrast in (28) in a neutral setting can be accounted for. Clearly, Heck & Himmelreich (2017) additionally would need to account for the fact that the contrasts in (2) and (28) are very similar.

Similar considerations apply to the contrast in (10). In (10a), the accusative object is realized to the left of the dative object. The examples in (10) differ from the examples in (25) in that both arguments are indefinites (which they must be to fit into Heck & Himmelreich's explanation). Once again assuming that the *wh*-phrase and the FQ form a constituent prior to the *wh*-phrase being relocated into Spec,C, there is just one linearization constraint at stake in (10), which is violated in (10a), and not violated in (10b).

4 Conclusion

The present paper has shown that neither the conceptual, nor the empirical argument presented in Heck & Himmelreich (2017) in favour of base positions of arguments in German clauses can be kept up against closer scrutiny. The conceptual argument depends on a specific architecture of grammar which makes use of a generator function only but does not apply linearization constraints. A model of grammar, in which both a generator function and violable linearization constraints are employed, can not only account for the contrasts discussed in Heck & Himmelreich (2017), but also for parallel data, which do not lend themselves to an analysis in terms of intervention, such as examples with focused and non-focused definite NPs (24), and cases of apparent scrambling within VP (25).

The empirical argument does not consider the crucial role of animacy. If animacy is taken into account, we arrive at the surprising result that two alternative orders become optional, which clearly goes against the assumption of a single base position. In addition, there are no

mechanisms in Heck & Himmelreich's analysis to account for the disappearance of an observed contrast, which can only depend on whether the stimulus is animate or not.

Altogether, we are careful not to argue that the arguments presented herein show that a variable base generation approach is the only tenable approach to analyze German clause structure. As it currently stands, we agree with Salzmann (2025) that decisive arguments for either position are hard to find.

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